INCH-POUND

AN780 Rev 7
4 August 2011
SUPERSEDING
AN780 Rev 6
23 July 1956

DETAIL SPECIFICATION SHEET

NIPPLE, UNION

Inactive for new design after 24 July 1998.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-6001.

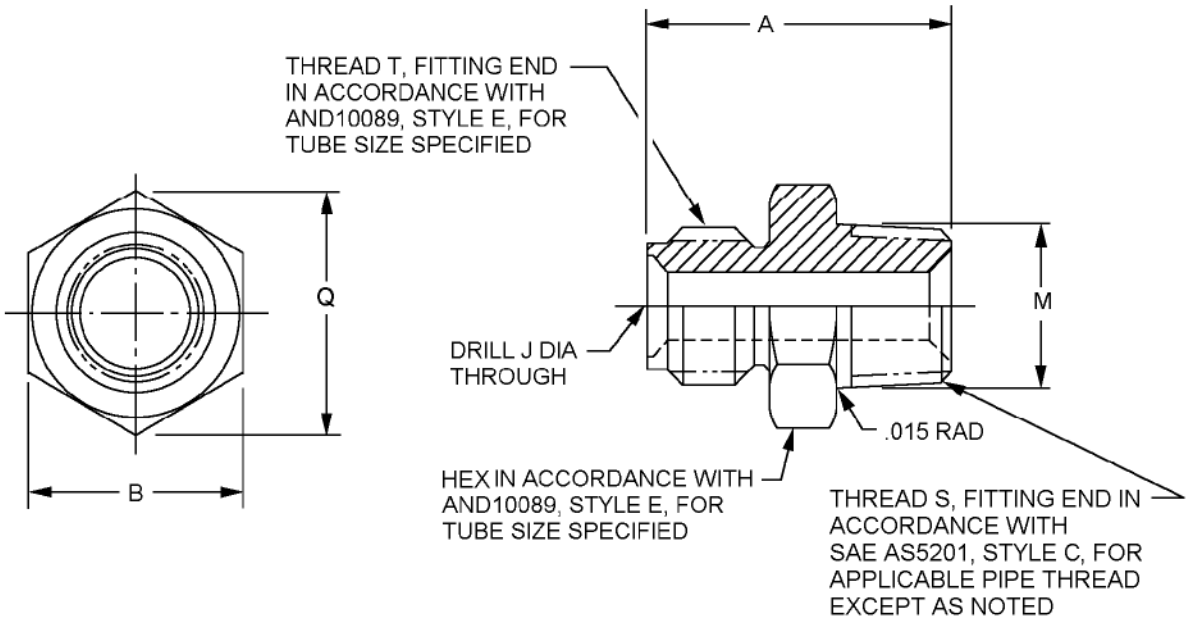


FIGURE 1. Nipple, union.

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TABLE I. Dash numbers and size. 1/ 2/

Dash number		Tubing outside diameter	Thread S SAE AS5201 ANPT	Thread T AND10089	A	B		J ±.005 Diameter	M Diameter	Q Reference	Weight Max	
Brass	Corrosion resistant steel										Brass lbs	Corrosion resistant steel
2	C2	.125	1/8-27	.3125-32 UNEF-3A	.812	.438	±.003	.125	.406	.51	.026	
3	C3	.188		.3750-24 UNF-3A	.938			.161			.028	
4	C4	.250		.4375-20 UNF-3A	1.000	.562	±.004	.189		.65	.038	
5	C5	.312		.5000-20 UNF-3A	1.062	.625				.72	.045	
6	C6	.375	1/4 - 18	.6250-18 UNF-3A	1.312	.750		.281	.547	.87	.080	
8	C8	.500	3/8 - 18	.7500-16 UNF-3A	1.500	.875		.406	.672	1.01	.125	
10	C10	.625	1/2 - 14	.9375-14 NS-3	1.750	.938		.531	.844	1.09	.185	
12		.750	3/4 - 14	1.1250-12 UNF-3A	1.875	1.250	±.005	.719	1.047	1.30	.270	
16		1.000		1.5000-12 UNF-3A		1.750				2.02		

TABLE I. Dash numbers and size – Continued. 1/ 2/

Dash number		Tubing outside diameter	Thread S SAE AS5201 ANPT	Thread T AND10089	A	B		J ±.127 Diameter	M Diameter	Q Reference	Weight Max	
Brass	Corrosion resistant steel										Brass Kg	Corrosion resistant steel
2	C2	3.175	1/8 - 27	.3125-32 UNEF-3A	20.625	11.125	±.0762	3.175	10.312	12.954	.012	
3	C3	4.775		.3750-24 UNF-3A	23.825			4.089			.013	
4	C4	6.350		.4375-20 UNF-3A	25.400	14.275	±.1016	4.801		16.510	.017	
5	C5	7.925		.5000-20 UNF-3A	26.975	15.875				18.288	.020	
6	C6	9.525	1/4 - 18	.6250-18 UNF-3A	33.325	19.050		7.137	13.894	22.098	.036	
8	C8	12.700	3/8 - 18	.7500-16 UNF-3A	38.100	22.225		10.312	17.069	25.654	.057	
10	C10	15.875	1/2 - 14	.9375-14 NS-3	44.450	23.825		13.487	16.358	27.686	.084	
12		19.050	3/4 - 14	1.1250-12 UNF-3A	47.625	31.750	±.1270	18.263	26.594	33.020	.122	
16		25.400		1.5000-12 UNF-3A		44.450				51.308		

1/ Dimensions are in inches
2/ Metric equivalents are given for information only.

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REQUIREMENTS

Dimensions. See table I. Unless otherwise specified tolerances on decimals are ±.010, angles ±0.5°.

Configuration. See figure 1.

Inactive for new design. Dash numbers 12 and 16 are inactive for new design.
Dash numbers 2, 3 and 4 are inactive for new design except for use in oxygen systems and engine primer lines.
Dash numbers 5, 6, 8 and 10 are inactive for new design except for use in oxygen systems.
Dash numbers 5, 6, 8 and 10 are not to be used on engine primer lines.

Materials: Brass; bars, shapes or forgings.
Corrosion – resistant steel; bars, shapes or forgings.
See procurement specification.

Finish: See procurement specification.

Identification of product. Add P before the dash number for cadmium – plated brass nipple.

AN780 X X
- X: Size
- X: Material, finish
- AN780: Drawing number

Example of Part or Identifying Number (PIN): AN780-4 nipple – union, brass, for .250 tubing OD.
AN780P4 nipple – union, brass, cadmium plated, for .250 tubing OD.
AN780C4 nipple – union, corrosion resistant steel, for .250 tubing OD.

Workmanship. Break all sharp edges and remove all hanging burrs and slivers which might become dislodged under usage.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Referenced documents. In addition to MIL-DTL-6001, this document references the following:

AND10089
SAE AS5201

CONCLUDING MATERIAL

Custodians:
Army – AR
Navy – SH
Air Force - 99
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2010-180)

Review activities:
Army – AT, AV, CR4
Navy – AS, CG, MC, SA, YD
Air Force - 71

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

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